Coalgebraic higher types and determinacy

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Abstract

Partial function types on numerical partial equivalence relations admit faithful coalgebraic realizations with countable carriers and binary observations. We compare this realization with restriction splitting and regular exact completion, and give an arithmetic-oracle bound for its finite-type fragment. A range obstruction occurs already at the type of partial numerical functions. These constructions and the obstruction hold in ZF. Determinacy supplies a different extension: strategies give Mealy or Moore reductions of stream predicates, and real determinacy gives continuous representative lifts on invariant comeager domains for quotients by countable group actions. These lifts have exact-domain asynchronous coalgebraic realizations. We identify the composition condition on their domains and a determinacy obstruction to complete real-valued invariants of numerical oracle presentations.

1 Computational types and the ambient set theory

A recursion category can occur as a category of computations inside a larger category of types. Restriction splitting supplies domains; exact completion supplies certain quotients. Function types require further quotients of programs. The question here is whether the resulting operations admit simultaneous coalgebraic realizations and which parts of that construction depend on choice.

We use ZF as the base theory. Additional axioms are stated in the results that use them. Write $\omega = \{0, 1, \ldots\}$ and $2 = \{0, 1\}$. Fix the Cantor pairing $\langle m, n\rangle = (m + n)(m + n + 1)/2 + n$ and its computable projections. Fix an acceptable numbering $(\varphi_e)_{e\in\omega}$ of the partial recursive numerical functions. Composition gf means first f, then g. Equality of partial maps includes equality of domains. The restriction $\bar{f}$ is the partial identity on $\operatorname{dom} f$.

A partial equivalence relation (PER) is a symmetric transitive relation $R \subseteq \omega^2$. Put $\operatorname{supp} R = \{a : aRa\}$ and $|R| = \operatorname{supp} R/R$. Let $\mathcal{P}$ have PERs as objects and partial functions $f : |R| \rightharpoonup |S|$ tracked by numerical programs as arrows. A tracker e must satisfy, for every $a \in \operatorname{supp} R$,

$$f([a]_R)\downarrow \iff \varphi_e(a)\downarrow, \qquad f([a]_R) = [\varphi_e(a)]_S \quad \text{when defined.}$$

In particular, convergence is invariant under R. Products use numerical pairing, and a singleton PER gives a restriction terminal object $\mathbf{1}$. Program composition and strict pairing make $\mathcal{P}$ a cartesian restriction category. Put $\mathcal{V} = \operatorname{Tot}(\mathcal{P})$, and let $\mathbf{N}$ be equality on ω. The full subcategory on positive finite powers of $\mathbf{N}$ is the numerical recursion category $\mathcal{R}$.

Define $R \Rightarrow_p S$ by

$$e\,(R \Rightarrow_p S)\,e' \quad\iff\quad \forall aRa' \begin{cases} \varphi_e(a), \varphi_{e'}(a') \text{ both diverge}, & \text{or} \\ \varphi_e(a), \varphi_{e'}(a') \text{ converge to } S\text{-related values.} \end{cases}$$

This is a PER whose classes are precisely the arrows $R \rightharpoonup S$. Universal evaluation descends to $\mathrm{ev} : (R \Rightarrow_p S) \times R \rightharpoonup S$. The parameter theorem gives total currying and the bijection

$$\mathcal{P}(Q \times R, S) \cong \mathcal{V}(Q, R \Rightarrow_p S). \tag{1}$$

Indeed, if e tracks h, a total compiler sends q to an index $s(e,q)$ satisfying $\varphi_{s(e,q)}(a) \simeq \varphi_e(\langle q, a\rangle)$. Input equivalence for h proves extensionality of this compiler. Evaluation proves the two inverse identities in (1). The bijection is natural under total changes of Q.

The computation PER LS identifies indices whose computations on input 0 both diverge or return S-related values. Suspension and forcing give $\mathcal{P}(R,S) \cong \mathcal{V}(R, LS)$. The unit returns an immediate computation, and multiplication runs an outer computation and then the computation it returns. These explicit compilers define a monad on $\mathcal{V}$, with Kleisli category $\mathcal{P}$.

Let $\mathcal{H}$ have finite type expressions generated from $\mathbf{1}, \mathbf{N}$ using products, L, and $\Rightarrow_p$ as objects, with all $\mathcal{P}$-arrows between their PER interpretations. Retaining the expressions as object labels makes changes of numbering unambiguous. This includes both strict function types and lazy types. For the latter, set $V_\iota = \mathbf{N}$, $C_\sigma = LV_\sigma$, and $V_{\sigma\to\tau} = C_\sigma \Rightarrow_p V_\tau$. A lazy function receives a computation name. Abstraction returns a function value; application forces the function computation and passes the argument name to it. The substitution theorem for these types is proved in [1, Section 6].

All these definitions are available in ZF. Every class of a numerical PER has a least member, so $|R|$ is canonically bijective with $\{a \in \operatorname{supp} R : \forall b < a\ \neg(bRa)\}$. This selection is set-theoretic; the evaluation and currying algorithms above do not require it to be computable.

2 Comparison with the existing completions

Let K_r denote splitting of restriction idempotents. Let $\mathcal{R}^1$ have the positive finite powers of ω and a singleton as objects, with all partial recursive maps between them. The singleton is its restriction terminal object. Set $\mathcal{K} = K_r(\mathcal{R}^1)$, $\mathcal{D} = \mathrm{Tot}(\mathcal{K})$, and $\mathcal{E} = \mathrm{Ex}_{\mathrm{reg}}(\mathcal{D})$. Here $\mathcal{D}$ is equivalent to the regular category of c.e. numerical supports and partial recursive maps total on their supports. Its regular exact completion has the presentation by c.e. PERs and c.e. saturated total functional relations described in [3, section "Completions of categories of partial operators"]. Write $\mathcal{P}_{\mathrm{ce}}$ for the full subcategory of $\mathcal{P}$ on c.e. PERs.

Proposition 2.1. *There are fully faithful functors $\mathcal{K} \to \mathcal{P}$ and $\mathcal{E} \to \mathcal{V}$. The first identifies split domains with equality PERs on c.e. supports. The second identifies $\mathcal{E}$ with the full subcategory of $\mathcal{V}$ on c.e. PERs. Moreover,*

$$\mathrm{Par}(\mathcal{E}, \mathrm{Mono}(\mathcal{E})) \simeq \mathcal{P}_{\mathrm{ce}}.$$

Proof. A restriction idempotent on a numerical object is a partial identity on a c.e. set. Its splitting therefore has precisely the indicated support and partial recursive arrows, proving the first assertion.

A c.e. total functional relation F between PERs R, S is executed by enumerating $F(a,b)$ until an output b appears. Totality ensures termination on valid inputs, and functionality ensures that all outputs belong to the same S-class. Conversely a total tracker e defines the saturated c.e. relation

$$F_e(a,b) \quad\Longleftrightarrow\quad aRa \ \wedge\ \exists c\,[\varphi_e(a)\downarrow = c \ \wedge\ cSb].$$

Extensionality supplies saturation in the first coordinate. These constructions identify the homsets, and represented functions determine identities and composition.

For a partial tracker, the name-domain $D_e = \{a : aRa \wedge \varphi_e(a)\downarrow\}$ is c.e. and R-saturated. Restricting R to D_e gives a domain span in $\mathcal{E}$. Conversely, consider a span $R \xleftarrow{m} T \xrightarrow{g} S$ with m

monic. Its underlying map on classes is injective: distinct classes are separated by maps from $\mathbf{1}$. On input a, search for a valid T-name whose m-image is R-related to a, and apply a tracker of g to it. The search is effective because the presentation is c.e. Monicity makes the output class independent of the chosen preimage. Its convergence domain is exactly the domain of the span. This proves realizability of every span. Applied to the mono m, the same search gives its inverse onto its c.e. saturated image in R. Thus the span is isomorphic to the reconstructed domain span, which also proves faithfulness of the partial-map comparison. □

This comparison does not identify all PERs with exact-completion objects. The support of $\mathbf{N} \Rightarrow_p \mathbf{N}$ is ω, but its relation is extensional equality of partial recursive functions and is not c.e. Its total-function subspace has the additional non-c.e. support $\mathrm{TOT} = \{e : \varphi_e \text{ is total}\}$. The obstruction in [3, section "Total function spaces and finite observations"] shows that $\mathcal{E}$ has no exponential of the numerical object by itself. The larger category supplies operations unavailable in that completion.

3 A simultaneous coalgebraic realization

Let $M(X) = 2 \times X^{\{+,-\}}$. An M-coalgebra is a set with a binary output and two transitions. Partial coalgebra morphisms have transition-closed subcoalgebra domains. Let Q have carrier $\mathbb{Z}$, output 1 precisely at $0, 1$, and transitions adding or subtracting 1. A coalgebra endomorphism of Q commutes with translations, so it is $n \mapsto n + k$. Preservation of the two marked positions forces $k = 0$. A nonempty subcoalgebra of Q is all of Q.

For a PER R, define $J(R)$ to be the coproduct of one copy of Q for each element of $|R|$:

$$J(R) = |R| \times \mathbb{Z}, \qquad o([a], n) = \chi_{\{0,1\}}(n), \qquad \delta_{\pm}([a], n) = ([a], n \pm 1).$$

For $f : R \rightharpoonup S$, define $Jf([a], n) = (f([a]), n)$ on $\mathrm{dom}\, f \times \mathbb{Z}$. This uses the rigid marker construction of [2, Section 4] with a single label.

Theorem 3.1. *In ZF, $J : \mathcal{P} \to \mathrm{Par}(\mathrm{Coalg}(M))$ is a faithful functor with countable carriers. It preserves restrictions, zeros, and binary near products. Its restriction to $\mathcal{H}$ realizes application, total currying, the computation monad, and lazy substitution simultaneously in the category of represented arrows.*

Proof. Every subcoalgebra of $J(R)$ is a union of whole copies of Q. A map between two such unions sends each copy to one target copy, and rigidity of Q forces it to preserve integer phase. Thus the construction $X \mapsto X \times \mathbb{Z}$, for arbitrary sets X, embeds **Pfn** fully faithfully into partial coalgebra maps. The functor J is the faithful underlying-set functor of $\mathcal{P}$ followed by this marker embedding. This proves functoriality, faithfulness, and preservation of restrictions and zeros. Least representatives code each $|R| \times \mathbb{Z}$ into ω.

The coalgebra product of $J(R)$ and $J(S)$ is $|R| \times |S| \times \mathbb{Z}$. Indeed, two maps from any coalgebra into the two factors assign the same phase to each source state: their binary outputs after all words in $+, -$ coincide, and these outputs determine the phase uniquely. The two maps therefore pair into the displayed coalgebra. The bijection $|R \times S| \cong |R| \times |S|$ gives the canonical product comparison, including the intersection of partial domains.

Every operation is an arrow of $\mathcal{P}$, so J sends its typed equations to equalities of partial coalgebra morphisms. Total arrows remain total. The represented category, with source objects retained as tags, is isomorphic to $\mathcal{P}$; hence (1) remains its representing bijection. The monad on its total arrows is $JR \mapsto J(LR)$, with unit and multiplication obtained by applying J. These statements

concern represented arrows. The terminal marker $Q = J\mathbf{1}$ is terminal on total maps in the marker subcategory, rather than the final coalgebra of all of $\mathrm{Coalg}(M)$. □

For example, after the product comparison, application is the concrete partial coalgebra morphism

$$(([e],[a]),n) \longmapsto ([\varphi_e(a)],n).$$

Its domain contains the entire integer orbit whenever the computation converges. Currying sends $([q],n)$ to $([s(e,q)],n)$, without executing the body. For lazy arguments, the $[a]$ coordinate is a class in a computation PER; the function decides whether to force it. Thus the construction retains both divergence and the multiplicity of states having identical observations. The final Moore behavior of $([a],n)$ depends only on n: the value $[a]$ is retained by its orbit copy. Application is realized by a morphism between these systems; the shifts themselves do not execute the application program.

3.1 An effective bound and oracle relatives

Write $0^{(\omega)} = \{\langle n,k\rangle : k \in 0^{(n)}\}$ for the join of the finite Turing jumps. It has the degree of arithmetical truth.

Proposition 3.2. *For finite type expressions defining $\mathcal{H}$, the coalgebra carriers, outputs, transitions, and represented arrow maps have presentations uniformly computable relative to $0^{(\omega)}$. For the category of partial A-recursive numerical maps, the same construction has bound $A^{(\omega)}$.*

Proof. Induction on a type expression produces an arithmetical formula for its PER. Products use pairing and conjunction; L and $\Rightarrow_p$ add numerical quantifiers and halting predicates. Arithmetical truth therefore decides support and equivalence uniformly across all finite types. It also decides the canonical-carrier condition $aRa \wedge \forall b < a\ \neg(bRa)$.

On a canonical input a, execute a supplied tracker e. If it returns b, choose the least $c \le b$ with cSb using the oracle. This computes the represented partial map with its exact domain. Integer shifts and marker outputs require no further oracle. All tests relativize to A. On the original numerical objects the PERs are equality, so canonicalization is trivial and the arrow action retains its original computability. □

Let φ, ψ be two acceptable numberings, and let $\mathcal{H}_\varphi, \mathcal{H}_\psi$ be their expression-labelled type categories. Changing the numbering gives a concrete comparison of their representations. The numerical translations are extended recursively to products and computation types. Write t_R, u_R for forward and backward name translations at type R. At $R \Rightarrow_p S$, compile in the new numbering the program $a \mapsto t_S(\varphi_e(u_R(a)))$. The parameter theorem produces this code totally. Induction gives quotient bijections b_R and preserves convergence. The equivalence $F : \mathcal{H}_\varphi \to \mathcal{H}_\psi$ is the identity on type expressions and sends $f : R \rightharpoonup S$ to $b_S f b_R^{-1}$. Its coalgebra comparison is

$$\kappa_R : J(R) \longrightarrow J'(FR), \qquad ([a]_R, n) \longmapsto ([t_R(a)]_{FR}, n)$$

is a coalgebra isomorphism. For every partial $f : R \rightharpoonup S$, the equality $J'(Ff)\kappa_R = \kappa_S Jf$ includes equality of domains. The same translations commute with evaluation, suspension, and currying. Write $c^P_{R,S} : P(R \times S) \to PR \times PS$ for a functor's canonical product comparison. Then

$$c^{J'F}_{R,S}\kappa_{R\times S} = (\kappa_R \times \kappa_S)c^J_{R,S},$$

since product names are translated componentwise. These are the comparison equations of [2, Section 3], applied to typed cartesian restriction categories. Restriction to the numerical recursion core gives a 1-cell in the representation 2-category defined there.

4 A range obstruction at the first function type

A Heller range of $f : X \rightharpoonup Y$ is a restriction idempotent r with $rf = f$ such that $uf = vf$ implies $ur = vr$ for all parallel $u, v : Y \rightharpoonup Z$. It is least among restriction idempotents d satisfying $df = f$: apply the condition to d and id_Y. The following obstruction involves a generated type, without adjoining an arbitrary additional PER.

Theorem 4.1. *Let $S = \mathbf{N} \Rightarrow_p \mathbf{N}$, and let $z : \mathbf{1} \to S$ select the total-zero function. In $\mathcal{H}$, z has no least supporting restriction idempotent and hence no Heller range. Moreover, z is not a partial isomorphism: there is no $s : S \rightharpoonup \mathbf{1}$ with $sz = \mathrm{id}_{\mathbf{1}}$ and $zs = \bar{s}$. These conclusions hold in ZF.*

Proof. For each n, a program tracking $r_n : S \rightharpoonup S$ runs its input index e on n, returning e if the result is 0 and otherwise diverging. This is an extensional restriction idempotent, and $r_n z = z$. If a least supporting restriction r existed, then $r \leq r_n$ for every n. Together with $rz = z$, this makes the numerical domain of every tracker of r exactly

$$Z = \{e : \forall n \; \varphi_e(n) \downarrow = 0\}.$$

This set is not c.e. Let $K = \{k : \varphi_k(k) \downarrow\}$. Uniformly in k, compile a program which, on input n, returns 0 if $\varphi_k(k)$ has not halted within n steps and diverges otherwise. Its index belongs to Z exactly when $k \notin K$. Thus a tracker of r would enumerate the non-c.e. complement of K.

A partial isomorphism inverse $s : S \rightharpoonup \mathbf{1}$ would satisfy $sz = \mathrm{id}_{\mathbf{1}}$ and $zs = \bar{s}$. Its domain would be precisely the zero-function class, giving the same contradiction. □

The theorem also rules out an exact realization of the singleton image by a tracked partial identity. It persists after adding DC, AD, or real determinacy: its proof uses only numerical computation and ZF.

Proposition 4.2. *The functor J preserves every Heller range that exists in the full PER category $\mathcal{P}$.*

Proof. Let r be a Heller range of $f : R \rightharpoonup S$ and put $I = \mathrm{im}\, f \subseteq |S|$. If I is empty, applying the range condition to id_S and 0 gives $r = 0$. Otherwise choose a name b with $[b]_S \in I$, set $B = \{a \in \mathrm{supp}\, S : [a]_S \in I\}$, and define the PER $T = S \cup (B \times B)$. The identity-name map $u : S \to T$ and the constant-b map $v : S \to T$ are total tracked maps. They satisfy $uf = vf$, so $ur = vr$. Since r is a partial identity, the latter equation forces $\mathrm{dom}\, r \subseteq I$; $rf = f$ gives the reverse inclusion. Thus r is the actual image restriction, and Jr selects exactly the image orbits of Jf. That partial identity is its ambient Heller range. □

The ranges of the original recursion category are also preserved by J. Dovetailing numerical computations until a preimage is found gives, for each f, a partial section s with $fs = \mathrm{ran}(f)$. In any restriction category, $r = fs$ and $rf = f$ imply $uf = vf \Rightarrow ur = ufs = vfs = vr$. A restriction functor preserves these equations. More generally, c.e. PER targets admit the c.e. saturated name-image of each arrow from a c.e. PER, so the comparison in Section 2 preserves their exact image ranges. The obstruction concerns the additional function type.

5 Dependent choice and countable coalgebras

Dependent choice (DC) states that every serial relation on a nonempty set admits a sequence following that relation from any specified starting point. A relation is serial when every element has a successor. Let $\mathcal{P}_+(X)$ denote the set of nonempty subsets of X.

Proposition 5.1. *Over ZF, DC is equivalent to the assertion that every pointed coalgebra $c : X \to \mathcal{P}_+(X)$ has an infinite run $x_0, x_1, \ldots$ satisfying $x_{n+1} \in c(x_n)$.*

Proof. Associate to c the serial relation $xRy \iff y \in c(x)$. Conversely, a serial relation defines this coalgebra by taking successor sets. The two assertions are identical under this correspondence. □

Deterministic transitions need no choice: their runs follow by recursion. On a carrier with a specified injection into ω, even nonempty successor sets have a least coded successor. DC becomes relevant for unrestricted state spaces or successively chosen lifts.

Countable Boolean algebras also admit the ultrafilters needed in countable germ representations without a choice axiom. For an enumeration $b_0, b_1, \ldots$ of a Boolean algebra and a proper ideal I, put $e_0 = 1$ and choose $e_{n+1} = e_n \wedge b_n$ if that meet is outside I; otherwise put $e_{n+1} = e_n \wedge \neg b_n$. The second meet is then outside I, since the join of the two meets is e_n. The set $U = \{b : \exists n\ (e_n \leq b)\}$ is an ultrafilter disjoint from I. Each decision is specified by a formula, so this is a ZF recursion. Applied to the proper ideals of agreement domains for unequal arrows, it gives the separating labels in the countable germ construction of [2, Section 4]. Coding a germ by a label index and an arrow index avoids a countable choice of fibre enumerations. This argument is restricted to explicitly countable presentations.

The Boolean Prime Ideal Theorem (BPI), in its ultrafilter formulation, asserts that every nontrivial Boolean algebra has an ultrafilter. It extends this separation argument to arbitrary Boolean algebras: apply it to the quotient by a proper ideal and pull back the ultrafilter. Applied to $\mathcal{P}(\omega)/\mathrm{Fin}$, it yields a nonprincipal ultrafilter on ω. Under AD every ultrafilter is countably complete [7, Theorem 2.8]; such an ultrafilter on ω would contain every cofinite tail and therefore their empty intersection. Thus BPI is incompatible with AD, while the countable separation argument above remains available in ZF.

6 Determinacy as a coalgebraic comparison theorem

Baire space ω^ω has the product topology of discrete ω. For a finite string s, let $[s]$ be its basic cylinder. A causal map $f : \omega^\omega \to \omega^\omega$ has $f(x)(n)$ determined by $x \restriction (n+1)$; a strictly causal map g has $g(y)(n)$ determined by $y \restriction n$. Causal maps have countable-state Mealy presentations, with output and transition map $X \times \omega \to \omega \times X$. Strictly causal maps have Moore presentations $X \to \omega \times X^\omega$. Each presentation includes a specified initial state. Finite input histories give the states in both constructions, with the empty history as initial state. Their tables can be coded as subsets of ω.

AD states that every length-ω perfect-information game with integer moves is determined. A winning strategy is a function on finite play histories, so it has a real code. PD restricts this assertion to projective payoff sets; here projective means obtained from Borel sets by finitely many projections and complements, with real parameters allowed.

Theorem 6.1. *Assume ZF+AD. For $A, B \subseteq \omega^\omega$, exactly one of the following alternatives holds:*
(i) a real-coded causal Mealy machine realizes f with $A = f^{-1}(B)$;
(ii) a real-coded strictly causal Moore machine realizes g with $\omega^\omega \setminus B = g^{-1}(A)$.
For projective A, B, PD suffices in place of AD.

Proof. Let I and II alternately produce $x(n)$ and $y(n)$, with II winning exactly when $x \in A \iff y \in B$. A winning II strategy produces $y(n)$ after seeing $x(n)$; its finite histories give the Mealy machine in (i). A winning I strategy produces $x(n)$ before seeing $y(n)$; its histories give the Moore

machine in (ii). Determinacy supplies one of these strategies. For projective A, B, the payoff is projective.

Both alternatives cannot hold. Their composite $h = gf$ is strictly causal and has a unique fixed stream. Construct its nth coordinate from the preceding coordinates, filling later coordinates temporarily with zero; strict causality makes the choice of that tail irrelevant. Induction proves existence and uniqueness of the fixed stream x. But then $x \in A \iff f(x) \in B \iff gf(x) \notin A$, a contradiction. □

This is the strategy proof of Wadge's lemma, with the two timing conventions made explicit [4, Theorem 7D.3]. The fixed stream is also obtained by feeding each Moore output back as the next input. For a Moore presentation (o, δ), its closed transition is $q \mapsto \delta_{o(q)}(q)$. For Borel payoff sets the game is already determined by Borel determinacy; AD extends the assertion to arbitrary stream predicates. The familiar Wadge order and its well-foundedness under determinacy are discussed in [5].

The real code of a strategy is an oracle for its machine, not a recursive program in general. A game requiring II to produce a fixed nonrecursive stream has a winning strategy and no recursive winning strategy. Thus determinacy gives additional real-coded comparisons without changing which numerical programs halt.

7 Representative lifts on invariant domains

Real determinacy, written $\mathrm{AD}_{\mathbb{R}}$, permits a whole real as each move. It implies AD. We use it with DC. A subset of a Polish space has the Baire property if it differs from an open set by a meager set; a set is meager if it is a countable union of nowhere dense sets, and comeager if its complement is meager. AD gives the Baire property to every set of reals [7, Theorem 2.7]. A map into Baire space whose basic-open preimages have this property is continuous on a comeager subset [6, p. 50]; the proof below chooses that subset to be invariant and G_δ.

Theorem 7.1. *Assume ZF+DC+*$\mathrm{AD}_{\mathbb{R}}$. *Let a specified countable group* Γ *act on* $X = \omega^\omega$ *by homeomorphisms, and let* E *be its orbit equivalence relation. Let* S *be an equivalence relation on a nonempty* $Y \subseteq \omega^\omega$. *Every function* $F : X/E \to Y/S$ *has a representative lift* $f : D \to Y$ *on an* E*-invariant comeager* G_δ *set* $D \subseteq X$, *with* f *continuous and*

$$[f(x)]_S = F([x]_E) \qquad (x \in D). \tag{2}$$

If instead ZF+DC+PD holds and the relation $P(x, y) \iff y \in Y \wedge F([x]_E) = [y]_S$ *is projective, the same conclusion holds.*

Proof. Every section of P is nonempty. In the real-move game where I first chooses x and II then chooses y, let II win iff $P(x, y)$; later moves are irrelevant. I has no winning strategy, since its first move admits a suitable reply. A winning II strategy therefore uniformizes P by a total function $f_0 : X \to Y$.

For each basic open set $U_n \subseteq \omega^\omega$, choose an open $V_n \subseteq X$ and a meager error set witnessing the Baire property of $f_0^{-1}(U_n)$. DC permits these countably many choices and their closed nowhere dense covers. Outside the union of all errors, enlarged to a meager F_σ set, the resulting comeager G_δ set G satisfies $f_0^{-1}(U_n) \cap G = V_n \cap G$ for every n. Hence $f_0|G$ is continuous.

Put $D = \bigcap_{\gamma \in \Gamma} \gamma^{-1} G$. Homeomorphisms preserve comeagerness, so the Baire category theorem makes D a dense comeager G_δ. The group law makes it invariant. Restricting f_0 to D proves (2).

Under PD, projective uniformization supplies a projective graph for f_0 [4, Theorem 6C.5]. Its basic-open preimages are projective and have the Baire property. The remaining argument is unchanged. □

Writing $q_D : D \to D/E$ and $q_Y : Y \to Y/S$ for the quotient maps, the equality just proved is the commutative square

$$\begin{array}{ccc} D & \xrightarrow{\quad f \quad} & Y \\ {\scriptstyle q_D}\downarrow & & \downarrow{\scriptstyle q_Y} \\ D/E & \xrightarrow[F|_{D/E}]{} & Y/S. \end{array}$$

The lift preserves output classes; it need not choose the same output name for equivalent inputs. Invariant uniformization requires that stronger equality of names, a distinction studied in [8].

7.1 An exact-domain machine for the lift

A pointed asynchronous transducer consists of a state set Q, an initial state $q_0 \in Q$, and a total transition and output rule $Q \times \omega \to (\omega \sqcup \{*\}) \times Q$, where $*$ means no output on that step. It defines a partial map on streams: the domain consists of inputs producing infinitely many numerical outputs, and the output stream omits the $*$ symbols.

Proposition 7.2. *Every continuous $f : D \to \omega^\omega$ with $D \subseteq \omega^\omega$ a dense G_δ has a real-coded countable-state asynchronous transducer whose partial stream function is exactly f, including its domain.*

Proof. Write $D = \bigcap_i O_i$ with each O_i open. A state records the input prefix and the number n of outputs already emitted. The initial state is the empty prefix with $n = 0$. After extending the input prefix to s, emit m precisely when

$$[s] \subseteq \bigcap_{i \le n} O_i \quad \text{and} \quad \forall x \in [s] \cap D \; f(x)(n) = m. \tag{3}$$

Otherwise emit $*$. Density of D ensures uniqueness of m. Advance n exactly when a numerical output is emitted.

For $x \in D$, continuity and openness imply that a sufficiently long prefix satisfies (3) for each successive n. Thus every output digit is eventually emitted and equals $f(x)(n)$. If $x \notin D$, choose j with $x \notin O_j$. No prefix of x lies inside O_j, so no output numbered j or later can be emitted. The productivity domain is exactly D. The rule is a function on a countable set of states and inputs; its numerical graph is a real code for the machine. □

Applied to Theorem 7.1, this gives a coalgebraic realizer of the quotient function on the saturated domain D/E. The transition table supplies an oracle for its execution. Because its productivity domain is G_δ, its name-level partial map is an arrow of the continuous-operator recursion category $\mathcal{B}_o$ of Montagna–Sorbi [9, Lemma 2.5, Definition 2.2, and Proposition 3.1].

7.2 Composition of the obtained partial maps

Fix Polish ambient spaces X_0, X_1, X_2. Regard the obtained maps as partial continuous maps $f : D \subseteq X_0 \to X_1$ and $g : E \subseteq X_1 \to X_2$, using the inclusion $Y \subseteq \omega^\omega$ for a lift's codomain. Their composite has domain $D \cap f^{-1}(E)$. It is again G_δ when D, E are G_δ in their respective ambient spaces. Quotient compatibility and saturation also compose. Comeagerness does not: take f

constant with value $y_0 \in X_1 = \omega^\omega$ and $E = \omega^\omega \setminus \{y_0\}$. Both original domains can be comeager while the composite has empty domain.

A sufficient condition is that f preserve category in the preimage sense: the inverse image of every meager subset of X_1 is meager in D. With D comeager in X_0, this makes $D \cap f^{-1}(E)$ comeager in X_0 whenever E is comeager in X_1. Such partial maps with comeager G_δ domains are closed under composition. The lifting theorem does not impose this condition, so its general conclusion belongs to the category of partial maps with actual G_δ domains.

8 Consequences for classification by integer sets

There is a distinction between giving each presentation a real code and assigning a complete real-valued invariant to its equivalence class. Determinacy imposes restrictions on the latter.

Under AD there is no transversal for eventual equality E_0 on 2^ω. To see this, suppose T chose one point per class. Its translates by finite coordinate flips are pairwise disjoint and cover Cantor space. All have the Baire property. If T is meager, this is a countable meager cover. If T is nonmeager, it is comeager in some cylinder. A nontrivial flip beyond that cylinder's specified coordinates makes a disjoint translate comeager in the same cylinder, again a contradiction. Thus quotient sets of reals behave differently from quotients of subsets of ω, whose least representatives exist in ZF.

Let $\mathcal{D}_T$ be the Turing degrees of subsets of ω. A cone is $\{d : d \geq_T b\}$ for some degree b. Under AD, the sets of degrees containing a cone form a countably complete ultrafilter [7, Theorems 2.8–2.9].

Proposition 8.1. *Assume ZF+AD. Every Turing-degree-invariant function $I : 2^\omega \to 2^\omega$ is constant on a Turing cone.*

Proof. For each n, the cone ultrafilter chooses one of the two sets of degrees on which $I(A)(n)$ is 0 or 1. Countable completeness makes the intersection of these chosen sets contain a cone. All coordinates of I are fixed on that cone. □

Let $\mathcal{R}_A$ have the positive finite powers of ω as objects and partial A-recursive functions as arrows. Then $\mathcal{R}_A = \mathcal{R}_B$ exactly when $A \equiv_T B$: simulation proves one direction, and the characteristic functions of the oracles prove the other. Equality here is equality as concrete subcategories of **Pfn**. There is therefore no complete real-valued invariant for equality of these presentations under AD. Indeed, a cone contains both a degree and its strictly higher jump, while Proposition 8.1 assigns them the same invariant. For classification under abstract or structured equivalence, the same obstruction applies if the corresponding categories do not all become equivalent on a cone. Whether they do is a separate categorical question.

9 The resulting extension problem

The numerical exact-completion comparison and the coalgebraic realization are established in Sections 2 and 3. They give higher-type operations and their comparison maps without choice. Theorem 4.1 prevents this particular program-quotient extension from having ranges for all its arrows. A completion retaining those ranges must change the admitted arrows or the interpretation of function types.

Determinacy supplies real-coded reductions of behavioral predicates. Real determinacy, or projective determinacy with projective data, also supplies representative lifts realized by partial continuous transducers. A higher-type construction based on those lifts must specify composable domains and compatible choices of realizers. Neither faithfulness of a coalgebra representation nor existence of individual lifts alone asserts a universal property for that construction.

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